Amar Tiwari

AI Engineer · Voice AI · Agentic Systems · RAG

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SUMMARY

AI Engineer shipping production LLM systems end to end: real-time voice agents on live phone lines, agentic research pipelines and offline-first RAG platforms. I own architecture, API and MCP integrations, tracing, evals and deployment across backend, web and mobile.

EXPERIENCE

AI Engineer 07/2026 – Present | Ahmedabad

Communication Crafts

- Built and shipped two production AI products end to end: Voicely (voice AI phone agents) and an AI lead-generation system, which I built solo from my 2nd-place internal Agentic AI Hackathon prototype (see Projects).
- Built a call-intelligence system for a client on my own: speaker-labelled transcription plus LLM extraction of sales objections (trigger, technique, outcome), shipped on Render and Vercel. Strong client feedback.

AI Engineer Intern 03/2026 – 06/2026 | Ahmedabad

Communication Crafts

- Shipped MarineDoc, a multi-tenant RAG SaaS, in one month with a 2-person team, and received a WOW Card for it (see Projects).
- Built EZ-Insights, an NL-to-SQL engine with schema RAG, a self-correcting retry loop, a 15-category SQL-injection validator and 55+ sensitive-column filters, for safe querying of live production databases.

PROJECTS

MarineDoc: Offline-First Multi-Tenant RAG Platform

FastAPI · Qdrant · BM25 · Supabase (Postgres) · Groq · React · React Native (Expo) · ONNX Runtime · sqlite-vec · Docker

- RAG SaaS for querying ship manuals with page-level citations in 3 modes: streamed LLM answers online, retrieval-only answers on the ship's intranet, and fully on-device RAG on mobile (ONNX + sqlite-vec) with no network.
- Hybrid retrieval: per-tenant dense + BM25 search, RRF fusion, two-stage cross-encoder reranking and parent-child expansion. PDF ingestion is cost-aware and only falls back from PyMuPDF to YOLOX layout analysis and OCR when needed.
- JWT tenant isolation, two-level RBAC, plans and quotas, a super-admin console with an audit log, RBAC-scoped delta sync to mobile, nightly reconciliation jobs and per-request trace IDs.

Voicely: Voice AI Agent Platform for Indian Telephony

Python · FastAPI · Pipecat · WebRTC · Twilio · Exotel · Sarvam · Deepgram · ElevenLabs · OpenAI · Next.js

- No-code platform for voice agents that make and answer real phone calls. Conversations are node graphs on a visual canvas, and each transition becomes an LLM function call.
- Real-time STT → LLM → TTS pipeline with swappable providers and 8 kHz audio end to end. Handles Hindi, Hinglish and 10 more Indian languages, including speakers switching languages mid-sentence.
- Mid-call HTTP tools, knowledge-base search, post-call data extraction, stereo recordings and timestamped transcripts. Safeguards: TRAI calling hours, signature checks, provider failover, and a carrier simulator for latency tests.

AI Lead Generation System

Python · FastAPI · Playwright · SQLite · React · Claude · Apollo API · Docker

- Scraper agent that collects posts from 10+ sources (APIs, RSS, JSON-LD, headless browser, IMAP alerts, Upwork GraphQL) and matches them against 245 service keywords. Respects robots.txt and pauses a source automatically when blocked.
- Claude research agents for each lead: research the company (Apollo + web), score it on a 100-point ICP model with cited evidence, find the decision maker, and draft a personalised email and LinkedIn note.
- React ops dashboard with live run tracing (every search, token and dollar per step), cost per sales-ready lead and usage analytics. Dockerised and deployed on Railway.

HACKATHON BUILDS

Cortex: 6-agent LangGraph SRE swarm across 2 AWS regions. CockroachDB is its only memory (vectors, lease locks), so it survives a region being killed mid-fix.

PipelineGuard: a DataHub MCP agent that generates dbt and Airflow code from real schema, lineage, PII tags and SLAs, then writes the lineage back.

StandupPilot: a Slack agent that uses Slack MCP and the Real-Time Search API to check reported blockers against live conversations.

SKILLS

Languages & Frameworks: Python, TypeScript / JavaScript, SQL · FastAPI, React, Next.js, React Native (Expo)

RAG & Retrieval: hybrid search (BM25 + dense, RRF), cross-encoder reranking, hierarchical chunking, OCR · Qdrant, Pinecone, FAISS, sqlite-vec

Data & Infra: PostgreSQL, Supabase, MongoDB, SQLite, WebSockets, SSE · Docker, AWS (EC2, Lambda, SAM, S3), Railway, Vercel, Render

LLMs & Agents: LangChain, LangGraph, LangSmith, MCP, tool calling, multi-agent orchestration, evals, guardrails · Claude, OpenAI, Groq, Gemini, Sarvam

Voice AI: Pipecat, real-time STT/TTS (Deepgram, Sarvam, ElevenLabs), WebRTC, Twilio, Exotel

EDUCATION

B.E. in Computer Science & Engineering 08/2022 – 05/2026
Government Engineering College, Patan (GTU) | CGPA: 8.01

ACHIEVEMENTS

2nd Place, Agentic AI Hackathon (internal), Communication Crafts, 2026

WOW Card as an intern for shipping MarineDoc in one month with a 2-person team

Bounty winner: "Level Up Your Aptos NFT Marketplace dApp"